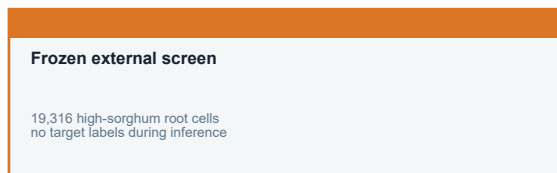
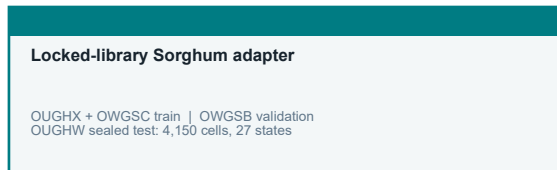


a Sealed-library adaptation restores external annotation

Frozen screen and label-supervised adaptation stay on separate evidence tiers



↓ LoRA, rank 8
3.03M trainable



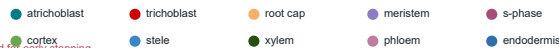
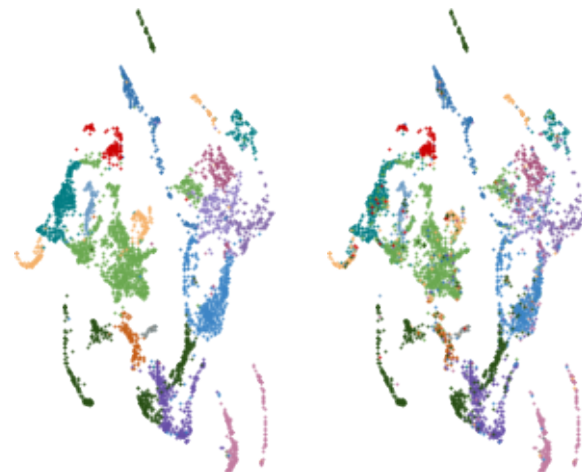
Target-species adaptation; not a zero-shot or third-party ranking. Raw 27-state accuracy 76.02%, macro-F1 0.7535; test labels never used for early stopping.

b Author topology is preserved in the sealed library

OUGHW only; each point is one held-out cell; color key is shared across author and prediction

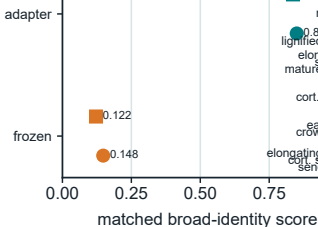
Author layer

Adapter layer



c Matched recovery

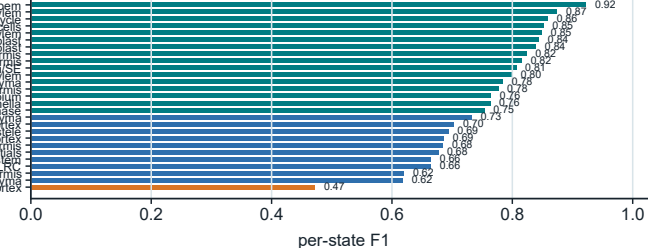
Points; 95% cell bootstrap



circles accuracy; squares macro-F1
3,549 same test cells

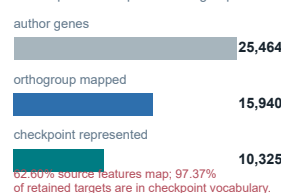
d Every author state remains visible

27-state OUGHW test



e Feature-transfer audit

Author-planned 10-species orthogroups



62.60% source features map; 97.37% of retained targets are in checkpoint vocabulary.